

Name:

Enrolment No:



MANIPAL UNIVERSITY
JAIPUR

Odd Semester End Term/Make Up Examination, January 2025

Faculty of Engineering
Department of Chemistry
B.Tech. Ist Year

Course Code: CHY1001/ CY1001 Course: Engineering Chemistry

Time: 03 hrs.

Semester: I
Max. Marks: 80

Instructions: All questions are compulsory.

Missing data, if any, may be assumed suitably.

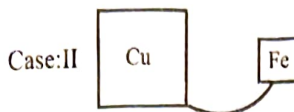
Calculator is allowed.

SECTION A (Memory Based Questions)

S.No.		Marks	CO
Q 1	Which of the following is not a natural polymer? (a) Rayon (b) Starch (c) Cellulose (d) RNA		1
Q 2	Distinguish between Chemical corrosion and Electrochemical corrosion.		1
Q 3	What are batteries? Give the types of batteries.		1
Q 4	The hardness of water is due to (a) K and Na salts (b) Salts of iron (c) Salts of Ca and Mg (d) SiO ₂	2 x 5 = 10	1
Q 5	Select a non-homogeneous system from the following. (a) Salt solution (b) Sugar solution (c) Glucose in water (d) Super-saturated solution of NaCl		1

SECTION B (Concept Based Questions)

Q 6	50 ml of standard hard water containing 1 gram of pure CaCO ₃ per liter consumed 20 ml of EDTA. 50 ml of hard water consumed 25 ml of same EDTA solution and EBT indicator. Calculate the total hardness of water sample in ppm.		4
Q 7	What is hydrogen-oxygen fuel cell? Discuss its working.		2
Q 8	Explain the mechanism of cationic polymerization reaction with suitable examples.	6 x 5 = 30	3
Q 9	What is principle of Boy's calorimeter? Draw a schematic diagram of Boy's calorimeter.		2
Q 10	Study the following two cases. The metals are connected and in presence of some electrolyte. Which metal will corrode? In which case rate of corrosion will be faster? Justify your answer. (the area of the squares indicates the relative surface areas of the metals)		5



SECTION-C (Analytical Based Questions)

Calculate the pilling Bedworth (P-B) ratio of the following metal oxides and find out which of the metal oxide coatings will be protective?

Q 11

Metal	Metal Oxide	Atomic weight of metal	Metal oxide density (g/cm^3)	Metal density (g/cm^3)
Zr	MgO	24.3	3.65	1.74
Sn	CaO	40.08	3.40	1.55
Na	Al ₂ O ₃	27.00	3.70	2.70

10×2
 $= 20$

Q 12

Draw and explain phase diagram of sulfur. Find the degree of freedom for each equilibrium system.

5

SECTION-D (Case Study/Application based Question)

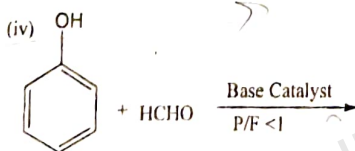
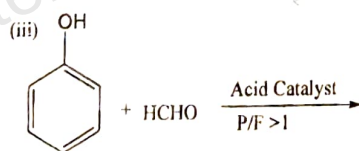
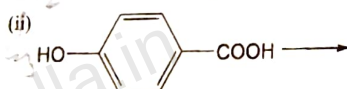
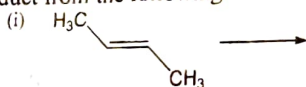
Q 13

Calculate the water equivalent value of a bomb calorimeter using following experimental data: Weight of benzoic acid = 2 g; weight of water = 2000 g; initial temperature = 25.0°C; final temperature = 30.0°C; fuse wire correction = 25 cal; cooling correction = 0.1°C. (calorific value of benzoic acid = 6325 cal/g)

4

Q 14

Predict the polymer product from the following:



10×2
 $= 20$

3